Advanced Plane Geometry

Section 1: Circle Theorems & Power of a Point

1 Two chords AB and CD intersect inside a circle at point P. If $AP = 4, PB = 9,$ and $CP = 3$, what is the length of PD?

a. 6

b. 8

c. 12

d. 15

2 From an external point P, tangent segment PT touches the circle at T, and secant line PAB passes through points A and B. If $PT = 10$ and $PA = 5$, what is the length of AB?

a. 10

b. 15

c. 20

d. 25

3 Two secants PAB and PCD meet outside a circle. If the intercepted arcs are $\overset{\frown}{BD} = 120^\circ$ and $\overset{\frown}{AC} = 40^\circ$, what is the measure of angle P?

a. 30°

b. 40°

c. 50°

d. 80°

Section 2: Triangle Centers & Cevians

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4 In $\triangle ABC$, medians intersect at the centroid G. If median AM has length 18 cm, what is the length of AG?

a. 6 cm

b. 9 cm

c. 12 cm

d. 15 cm

5 Which triangle center is equidistant from all three sides of a triangle?

a. Incenter

b. Circumcenter

c. Centroid

d. Orthocenter

6 In $\triangle ABC$, the circumcenter lies on the midpoint of the longest side. What type of triangle is $\triangle ABC$?

a. Acute

b. Right

c. Obtuse

d. Equilateral

Section 3: Solid Geometry & Volume Formulas

7 A right circular cone has a base radius of 6 cm and a slant height of 10 cm. What is the total volume of the cone?

a. 96π cm^3

b. 120π cm^3

c. 288π cm^3

d. 72π cm^3

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8 A solid metal sphere of radius 3 cm is melted down and recast into a cylinder of base radius 3 cm. What is the height of the cylinder?

a. 3 cm

b. 4 cm

c. 6 cm

d. 8 cm

9 If the linear dimensions of a rectangular prism are all multiplied by 3, by what factor does its surface area increase?

a. 3

b. 6

c. 9

d. 27

Section 4: Cyclic Quadrilaterals & Ptolemy's Theorem

10 In a cyclic quadrilateral $ABCD$, opposite angles are supplementary. If $\angle A = (4x + 10)^\circ$ and $\angle C = (3x + 30)^\circ$, what is the value of x?

a. 20

b. 25

c. 30

d. 35

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11 In cyclic quadrilateral $ABCD$, sides are $AB = 2, BC = 6, CD = 7,$ and $DA = 3$. What is the product of the diagonals $AC \times BD$ by Ptolemy's Theorem?

a. 32

b. 35

c. 39

d. 44

12 What is Brahmagupta's formula for the area of a cyclic quadrilateral with side lengths a, b, c, d and semiperimeter s?

a. $\sqrt{(s-a)(s-b)(s-c)(s-d)}$

b. $\frac{1}{2}(ac + bd)$

c. $\sqrt{s(s-a)(s-b)(s-c)}$

d. $\frac{ab+cd}{2}$

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